Augmenting Dysarthric Speech Severity Assessment with MOS Supervision

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Abstract

Dysarthria is a speech disorder marked by reduced intelligibility and communicative effectiveness. Automatic dysarthria assessment scales up support for diagnosing and treating neurological conditions such as Parkinson's disease. Yet training such systems is bottlenecked by the scarcity of clinically annotated dysarthric speech. This work proposes to augment dysarthric speech assessment using data from speech synthesis evaluations, specifically human-annotated utterances with Mean Opinion Score (MOS) labels from the QualiSpeech corpus. Experiments show that fine-tuning on speech synthesis assessment data consistently improves performance on both intelligibility and naturalness prediction, while joint training yields gains primarily on naturalness. These results suggest that synthesis artifacts and dysarthric speech share perceptual commonalities, and that speech synthesis evaluation corpora offer a practical augmentation source that reduces reliance on scarce clinical annotations.

Index Terms: Dysarthria, Dysarthric Speech, Automatic Dysarthria Assessment, Mean Opinion Score

1. Introduction

Dysarthria is a neuro-motor speech disorder resulting from neurological injuries or diseases, such as cerebral palsy, amyotrophic lateral sclerosis, Parkinson's disease, or stroke [1, 2]. These disruptions manifest in several perceptually salient characteristics, with reduced intelligibility and degraded naturalness being among the most prominent and impactful on daily communication.

Accurate assessment of dysarthric speech carries significant clinical and technological implications. It serves as a sensitive biomarker for early detection and tracking of neurological disease progression [3], provides an objective measure to guide speech therapy and monitor rehabilitation outcomes [4], and is a key factor for improving downstream systems such as automatic speech recognition (ASR) and disordered speech reconstruction [5, 6, 7, 8].

Despite its significant importance, clinical assessment of utterance-level intelligibility and naturalness still relies heavily on subjective evaluations by certified speech pathologists [9, 10], which are labor-intensive and difficult to scale. These limitations have motivated the development of automated, objective assessment methods. However, patients with speech disorders often present with co-occurring physical disabilities, making large-scale data collection particularly challenging. Furthermore, existing approaches are typically restricted to constrained lexicons and require matched control groups, resulting in an unnatural evaluation protocol. To the best of our knowledge, spontaneous utterances with unconstrained vocabulary have not been employed in prior studies.

To address the data scarcity issue, self-supervised learning (SSL) has emerged as a powerful paradigm in speech processing, offering robust and transferable representations from large-scale unlabeled corpora. Models such as wav2vec 2.0 [11] and HuBERT [12, 13] have been successfully applied to both detection and severity assessment of dysarthria [14, 15]. A wide spectrum of data augmentation techniques has also been explored, including signal processing [16], adversarial domain adaptation [17], generative approaches [18, 19, 20], reverse autoencoders that transform healthy speech into dysarthric speech [21], and other pre-trained TTS systems [22, 23, 24]. However, these generative approaches are designed to model spectro-temporal characteristics of dysarthric speech for ASR or speech reconstruction purposes. The augmented data they produce lacks perceptual validation, as no clinical experts re-annotate the generated samples with severity labels, leaving it unclear whether the synthetic speech aligns with human perceptual judgments. As a result, such data cannot be directly used as augmentation for severity assessment, where label quality and perceptual alignment are critical.

Meanwhile, automated assessment of Text-to-Speech (TTS) synthesis quality has been extensively investigated [25, 26], producing large quantities of human-annotated utterances with mean opinion score (MOS) labels measuring naturalness and intelligibility. Crucially, both synthesis artifacts and dysarthric speech represent deviations from natural human production, sharing perceptual characteristics such as reduced intelligibility, unnatural prosody, and degraded fluency. This acoustic and perceptual overlap motivates a novel data augmentation strategy, leveraging TTS assessment corpora, which are readily available and richly annotated, as an augmentation source for dysarthric speech assessment systems.

The contributions of this work are threefold:

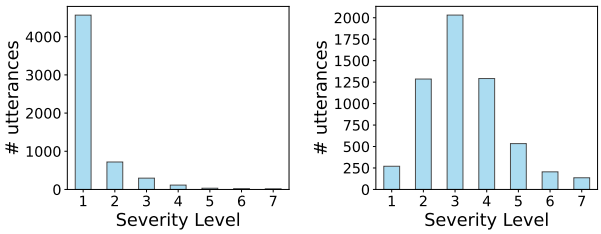
- An empirical study of SSL-based automatic dysarthric speech assessment on utterances with unconstrained vocabulary, going beyond the constrained protocols of prior work.
- Demonstration that synthesized speech with human-annotated MOS scores from TTS evaluation corpora can effectively augment dysarthric speech assessment systems.
- Evidence that synthesis failures and dysarthric manifestations share perceptual and acoustic commonalities, offering a novel perspective on cross-domain knowledge transfer between speech generation and articulatory disorders.

The rest of this paper is organized as follows. Section 2 introduces the assessment task and describes the SAP and QualiSpeech corpora. Section 3 presents the model architecture and the two training paradigms: joint training and fine-tuning. Sec-

tion 4 reports experiments and results on intelligibility and naturalness prediction. The last section concludes this work.

2. Task Description

2.1. The Speech Accessibility Project Challenge 2025



(a) *Intelligibility* (b) *Naturalness*

Figure 1: *Distribution of utterances across severity levels for* ***(a)*** *Intelligibility and* ***(b)*** *Naturalness in the Speech Accessibility Project (SAP) challenge. Intelligibility is heavily biased toward level 1 (minimal impairment), while Naturalness exhibits a more balanced spread across levels, reflecting the greater variability in perceived speech quality among dysarthric speakers.*

The Speech Accessibility Project (SAP) [27] challenge provides a large-scale, open-domain corpus of dysarthric speech. The corpus comprises utterances recorded from over 500 speakers diagnosed with Parkinson's disease, Down syndrome, amyotrophic lateral sclerosis, Cerebral palsy, or stroke, encompassing more than 400 hours of speech and over 190,000 utterances. The dataset was recorded through participants using their personal devices at home, capturing both read and spontaneously generated speech with unconstrained vocabulary to ensure naturalness and variability. Fine-grained perceptual speech assessment across multiple clinically relevant dimensions is provided in the corpus, including but not limited to, harsh voice, inappropriate silences, pitch level, naturalness, variable rate, and intelligibility. Each dimension is annotated by certified speech-language pathologists using a 7-point scale, where a higher score indicates more significant severity is manifested in the corresponding dimension. Given the complexity among these perceptual dimensions and the primary objective of dysarthric speech assessment, the focus is specifically narrowed to two core utterance-level annotations, Intelligibility and Naturalness. Distribution of utterances across severity levels can be seen from Figure 1.

Due to the limited availability of the original SAP test set, the development set of the corpus was repurposed as the test set. For each dimension, the training set comprises utterances specifically annotated for that dimension; accordingly, for Naturalness, 5,040 training utterances were obtained, while for Intelligibility, 5,046 utterances were selected. Validation sets consisting of 500 utterances each were randomly sampled from the corresponding training subsets. The resulting test sets include 714 utterances for Naturalness and 716 utterances for Intelligibility.

2.2. QualiSpeech: A Descriptive Corpus for Speech Quality Assessment

QualiSpeech [28] is an English corpus for low-level speech quality assessment with comprehensive annotations and detailed descriptive comments. It comprises three speech categories: synthetic speech (49%), primarily from BVCC [29] and including various TTS models with standardized MOS scores; simulated real speech (27%), sourced from NISQA [30] featuring distortions that emulate actual transmission; and real speech (24%), encompassing NISQA live recordings and in-the-wild samples from GigaSpeech [31]. To ensure balanced speech quality conditions, 20% of the synthetic speech samples are mixed with noise at signal-to-noise ratios ranging between 0 and 15 dB, while real speech samples are stratified into four quality groups based on UTMOS-predicted MOS scores [32]. Each utterance is evaluated across seven perceptual quality dimensions, namely noise, distortion, speed, continuity, listening effort, naturalness, and overall quality. This protocol results in training, validation, and test splits of 10,558, 2,167, and 1,852 utterances, respectively, with balanced compositions of synthetic and real speech.

Among the seven annotated dimensions, overall quality and naturalness are selected as they provide complementary yet comprehensive perspectives on perceptual evaluation. Overall quality captures the aggregated listener impression across multiple degradations, while naturalness reflects the degree to which speech resembles genuine human production. Focusing on these two dimensions ensures both clinical relevance for pathological speech assessment and consistency with established practices in general speech quality evaluation.

Table 1: *Details of self-supervised learning (SSL) pre-trained encoders.*

Model	# Param.	Dataset
wav2vec 2.0 Base[1]	94M	Librispeech [33]
wav2vec 2.0 Large*[2]	315M	Libri-Light [34]
wav2vec 2.0 Large+[3]	315M	Libri-Light [34] + CommonVoice [35] + Switchboard [36] + Fisher [37]
HuBERT Base[4]	95M	Librispeech [33]
HuBERT Large[5]	316M	Libri-Light [34]

3. Method

3.1. Model Architecture

To leverage transferable representations from large-scale unlabeled speech, self-supervised learning (SSL) pre-trained encoders are adopted as the backbone for feature extraction [38]. The core idea is to augment the limited in-domain dysarthric speech data with human-annotated TTS assessment data from QualiSpeech, using either joint training or fine-tuning to transfer perceptual supervision into the dysarthria assessment model. Given a raw waveform input, the SSL encoder produces a sequence of frame-level contextual representations. These are aggregated via mean pooling over the time dimension to obtain a fixed-dimensional utterance-level embedding, which captures global perceptual characteristics without requiring explicit seg-

[1] https://dl.fbaipublicfiles.com/fairseq/wav2vec/wav2vec_small.pt
[2] https://dl.fbaipublicfiles.com/fairseq/wav2vec/wav2vec_vox_new.pt
[3] https://dl.fbaipublicfiles.com/fairseq/wav2vec/w2v_large_lv_fsh_swbd_cv.pt
[4] https://dl.fbaipublicfiles.com/hubert/hubert_base_ls960.pt
[5] https://dl.fbaipublicfiles.com/hubert/hubert_large_ll60k.pt

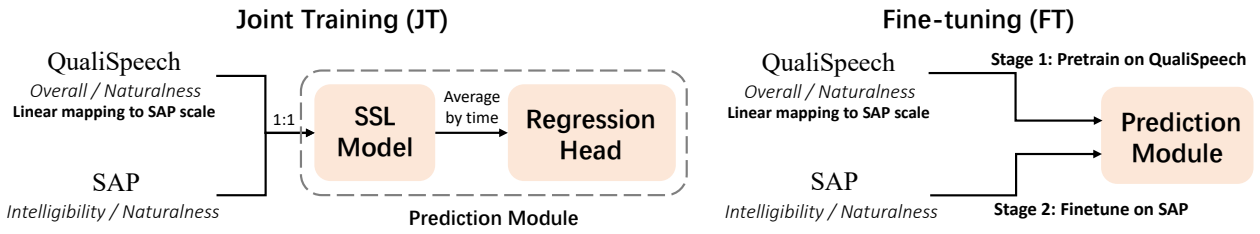


Figure 2: *An illustration of the proposed training paradigms. In Joint Training (JT), QualiSpeech and SAP utterances are mixed at a 1:1 ratio and fed jointly into a shared SSL encoder and regression head. In Fine-Tuning (FT), the model is first pre-trained on QualiSpeech, then fine-tuned on SAP. Both paradigms use linear scale mapping to align QualiSpeech MOS scores with the SAP severity scale.*

mentation. The utterance embedding is then passed to a regression head consisting of a two-layer feed-forward network with ReLU activation and dropout regularization between layers, producing a single continuous severity score. The entire model, including the SSL encoder, is fine-tuned in an end-to-end fashion during training.

3.2. Training Paradigms

Two training paradigms are proposed to incorporate perceptual supervision from QualiSpeech into dysarthric speech assessment, as illustrated in Figure 2.

Joint Training (JT). The JT paradigm trains a single model simultaneously on both the QualiSpeech and SAP corpora under a unified regression objective. To prevent the larger QualiSpeech corpus from dominating training, utterances are randomly sampled from the QualiSpeech training set to match the size of the SAP training split, yielding a balanced 1:1 mixture ratio. Since the two datasets adopt different rating scales, with QualiSpeech using a MOS scale of 1–5 while SAP uses a severity scale of 1–7, a linear transformation is applied to align QualiSpeech scores before joint optimization:

$$\hat{s} = 1 + (5 - s_{\text{MOS}}) \cdot \frac{6}{4}, \tag{1}$$

where $s_{\text{MOS}} \in [1, 5]$ is the original QualiSpeech MOS and $\hat{s}$ is the aligned score in the SAP scale. Under this mapping, a MOS of 5 (highest quality) corresponds to an SAP score of 1 (no dysarthric characteristics), and a MOS of 1 (lowest quality) maps to an SAP score of 7 (most severe).

Fine-Tuning (FT). The FT paradigm decouples the two stages of learning. The model is first pre-trained on QualiSpeech to acquire perceptual quality representations, then fine-tuned on the SAP corpus for dysarthric severity prediction. This sequential approach allows the model to initialize from a perceptually informed state before adapting to the clinical domain, without requiring explicit score alignment between datasets.

3.3. Evaluation Metrics

Model performance is evaluated using three complementary metrics: Mean Squared Error (MSE), Linear Correlation Coefficient (LCC), and Spearman's Rank Correlation Coefficient (SRCC). MSE quantifies the absolute deviation between predicted scores and ground-truth ratings, with lower values indicating higher prediction accuracy. LCC measures the strength of the linear relationship between predictions and references, while SRCC assesses the consistency of relative ranking order, which is particularly relevant for severity-level discrimination. Together, these three metrics provide a comprehensive view of both regression accuracy and perceptual alignment with human judgments.

4. Experiments and Results

To leverage human-perception-aligned supervision from QualiSpeech for dysarthria assessment, we evaluate two TTS-based data augmentation paradigms: joint training (JT) and fine-tuning (FT). All models evaluated are optimized using the Adam optimizer with mean squared error (MSE) as the training objective, a learning rate of $1e-5$, and weight decay of 0.01. Details of the utilized self-supervised learning (SSL) pre-trained encoders are listed in Table 1. All training paradigms share an identical experimental setup with parameters of SSL encoders kept tuneable. Under the JT paradigm, 4,000 utterances are randomly sampled from the QualiSpeech training set with the original data distribution preserved, and then combined with the SAP training set for joint optimization.

4.1. Performance on the Intelligibility Dimension

Table 2 (Row 1–5) reports intelligibility prediction on SAP, with Overall quality and Naturalness from QualiSpeech used as auxiliary supervision under fine-tuning (FT) and joint training (JT) across five SSL encoders. Three key trends emerge:
(i) Under in-domain training (IDT, Row 1), wav2vec 2.0 Base achieves the strongest performance, with a 36.4% relative MSE reduction over Large+, suggesting that smaller encoders generalize better to this task without additional supervision.
(ii) Speech synthesis augmentation via FT (Row 2 vs. 1) consistently improves intelligibility prediction across all wav2vec encoders, with the largest relative MSE gain of 43.7% for Large+ and a 19.6% LCC improvement for Base, demonstrating that QualiSpeech overall quality provides effective perceptual transfer for intelligibility.
(iii) FT consistently outperforms JT across all encoder and dimension combinations (Row 2 vs. 3, Row 4 vs. 5), with Overall quality yielding the most robust MSE and LCC gains while Naturalness further improves SRCC, particularly for larger encoders, indicating that the choice of augmentation dimension interacts with model capacity.

4.2. Performance on Naturalness Dimension

Table 2 (Row 6–10) reports naturalness prediction on SAP, utilizing Overall quality and Naturalness from QualiSpeech as auxiliary supervision under fine-tuning (FT) and joint training (JT) across five SSL encoders. Three key trends emerge:
(i) Speech synthesis augmentation via FT (Row 7 vs. 6) consistently improves naturalness prediction across all encoders, with

Table 2: *Results of self-supervised learning (SSL) pre-trained encoders under joint training (JT) and fine-tuning (FT) on the Speech Accessibility Project dysarthric speech corpus (SAP) and QualiSpeech. The two "Dimension" columns denote the SAP target dimension (left) and the QualiSpeech auxiliary supervision dimension (right). "IDT" denotes in-domain training on SAP only, with no QualiSpeech augmentation ("–"). For each SSL encoder, Mean Squared Error (MSE ↓), Linear Correlation Coefficient (LCC ↑), and Spearman's Rank Correlation Coefficient (SRCC ↑) are reported.* ***Bold values*** *indicate the best result per encoder within each SAP dimension group.*

Dimension		**Method**	wav2vec 2.0 Base			wav2vec 2.0 Large*			wav2vec 2.0 Large+			HuBERT Base			HuBERT Large		
SAP	QualiSpeech		MSE	LCC	SRCC	MSE	LCC	SRCC	MSE	LCC	SRCC	MSE	LCC	SRCC	MSE	LCC	SRCC
Intelligibility	–	IDT	0.348	0.628	**0.482**	0.421	0.523	0.368	0.547	0.471	0.322	0.461	0.540	**0.406**	0.475	0.485	0.404
	Overall quality	FT	**0.272**	**0.751**	0.427	**0.351**	**0.612**	0.393	**0.308**	0.534	0.285	0.487	**0.594**	0.368	0.431	0.596	0.356
		JT	0.408	0.590	0.446	0.560	0.317	0.330	0.627	0.225	0.302	0.612	0.387	0.317	0.526	0.479	0.363
	Naturalness	FT	0.303	0.648	0.464	0.387	0.572	**0.401**	0.495	**0.566**	**0.443**	**0.379**	0.451	0.295	**0.379**	**0.608**	**0.464**
		JT	0.433	0.602	0.475	0.604	0.383	0.352	0.528	0.445	0.348	0.489	0.515	0.309	0.451	0.551	0.391
Naturalness	–	IDT	1.127	0.581	0.591	1.354	0.469	0.481	1.119	0.574	0.607	1.169	0.546	0.521	0.941	0.570	0.503
	Overall quality	FT	1.053	**0.718**	**0.723**	1.033	0.695	0.706	1.000	0.686	**0.696**	**0.847**	0.644	0.587	**0.819**	**0.701**	**0.690**
		JT	0.909	0.691	0.680	1.027	0.606	0.613	1.106	0.589	0.617	1.050	0.614	0.634	0.930	0.661	0.668
	Naturalness	FT	**0.717**	0.717	0.657	**0.800**	**0.709**	**0.713**	**0.880**	**0.692**	0.690	1.075	**0.646**	**0.635**	0.823	0.678	0.675
		JT	1.022	0.637	0.643	1.085	0.629	0.648	1.022	0.637	0.642	1.121	0.586	0.590	1.036	0.638	0.635

wav2vec 2.0 Large* showing the largest LCC (48.2% relative) and SRCC (46.8% relative) gains, confirming that overall quality from QualiSpeech transfers effectively to dysarthric naturalness assessment.

(ii) Unlike intelligibility, JT also yields consistent gains over IDT on naturalness (Row 8 vs. 6), with wav2vec 2.0 Base and Large* reducing MSE by 19.4% and 24.2% respectively, suggesting that the perceptual alignment between QualiSpeech quality ratings and dysarthric naturalness is sufficient to support joint optimization.

(iii) Using QualiSpeech Naturalness as auxiliary supervision under FT (Row 9) achieves the lowest MSE across all conditions, with wav2vec 2.0 Base and Large* achieving 36.4% and 40.9% relative MSE reductions, indicating that dimension-matched supervision yields the strongest transfer.

Taken together, these results demonstrate that the perceptual overlap between speech synthesis quality and dysarthric naturalness is sufficient to support effective cross-domain augmentation under both FT and JT, validating the use of speech synthesis evaluation corpora as a practical and label-efficient augmentation source for dysarthric speech assessment.

4.3. Analysis on the Performance Gap between FT and JT

The consistent gap between FT and JT on intelligibility can be attributed to cross-domain label misalignment and negative transfer under joint optimization. The QualiSpeech supervision signals, overall quality and naturalness, are semantically misaligned with SAP intelligibility: while the latter reflects clinical judgments of phonemic clarity and word-level recoverability, the former capture aggregated listener impressions of multidimensional degradations and human-likeness, which are more closely related to the naturalness perceptual axis than to articulatory intelligibility.

Under JT, a shared regression head and a unified MSE objective are applied over both corpora, implicitly treating the linearly mapped QualiSpeech scores as a direct proxy for SAP intelligibility severity. This assumption fails for intelligibility, as gradients from QualiSpeech continuously bias the shared representations toward explaining MOS variance, suppressing the discriminative acoustic cues required for intelligibility assessment.

FT sidesteps this conflict by decoupling the two learning stages: QualiSpeech pre-training initializes the encoder with perceptually informed representations, while subsequent fine-tuning on SAP allows the model to re-weight features toward the target domain without cross-dataset gradient interference, explaining its consistent advantage over JT. These findings suggest that speech synthesis evaluation data can serve as a viable augmentation source for dysarthric intelligibility assessment, provided that the transfer is mediated through sequential fine-tuning rather than joint optimization.

5. Conclusion

This study demonstrated that leveraging perceptual annotations from the QualiSpeech corpus significantly enhances automatic dysarthria assessment. Fine-tuning consistently improved prediction accuracy over the baseline on both dimensions, while joint training yielded gains primarily on naturalness. Larger models derived greater benefit from additional supervision, and naturalness prediction consistently achieved stronger correlation with human judgments than intelligibility, likely attributable to the severe class imbalance in the intelligibility dimension as shown in Figure 1. The effectiveness of synthetic speech augmentation underscores perceptual and acoustic commonalities between synthesis failures and dysarthric speech, highlighting promising directions for future clinical research.

6. Generative AI Use Disclosure

During the preparation of this manuscript, the authors used generative AI to correct grammar mistakes and misspellings. After using this tool, the authors carefully reviewed and edited the manuscript, and take full responsibility for the final content of the paper.

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